Onyero Walter Ofuzim

Calgary, AB, Canada — Open to relocate across Canada — Available immediately

Email | Phone | GitHub | Portfolio | LinkedIn | ResearchGate

Role Fit Matrix

Role Family	Fit Estimate	Why the Fit is Reasonable
Control Systems / System Identification / Battery Modelling	90–95%	Strongest fit. Direct M.Sc. research in SPMe-inspired battery modelling, nonlinear system identification, estimation, optimization, and scientific Python.
Data Engineering / Analytics Engineering	78–85%	Strong Python, SQL, PySpark, lakehouse, Databricks, Delta Lake, dbt, Airflow/Prefect, Power BI, and portfolio projects. Industry experience supports KPI analytics and reporting.
Software Engineering / Backend / API Development	72–80%	Good project coverage in FastAPI, Java Spring Boot, ASP.NET Core, REST APIs, testing, CI, Docker, and system design. Less full-time production software experience than pure SWE candidates.
Machine Learning / Applied AI / MLOps	70–78%	Good applied ML, TensorFlow/PyTorch, scikit-learn, RAG, model evaluation, and research exposure. Stronger for applied/engineering ML than research scientist roles.
DevOps / Cloud / Platform Engineering	65–72%	Solid project exposure with Docker, Kubernetes basics, OpenCost, Grafana, Kyverno, GitHub Actions, Terraform basics, and observability. Better suited to junior/entry platform roles.
Network Engineering / NOC / Telecom Analytics	82–88%	Strong MTN network operations background, SNMP, KPI monitoring, troubleshooting, Excel/Power BI reporting, Linux/Bash, and automation.
Embedded / IoT / Electrical Engineering	70–78%	Relevant IoT battery monitoring, PIC/Arduino exposure, electrical engineering background, control systems, and instrumentation awareness. Fit depends on depth of C/C++/RTOS requirement.
Robotics / Autonomy	62–70%	Transferable control, modelling, estimation, and simulation experience, plus foundational robotics exposure. Better fit for controls/autonomy-adjacent roles than pure robotics software roles.
Quantitative / Financial Engineering	60–70%	Current WQU financial engineering studies, Python/statistics background, and quantitative projects. Stronger after adding more finance-specific projects.
Technical Consulting / Solutions Engineering	78–85%	Broad engineering, software, analytics, stakeholder communication, teaching, documentation, and demo-building experience. Good fit for technical client-facing roles.

Education

M.Sc., Electrical & Software Engineering Jan 2024 – May 2026
University of Calgary, Calgary, AB
GPA: 3.85
Focus: Control Systems, System Identification, Battery Modelling, Optimization, Scientific Computing

- Department of Electrical & Software Engineering Funding 2024
- International Graduate Tuition Award 2024
- Dept./Faculty Scholarship Award, Exceptional Student 2024

M.Sc., Financial Engineering 2025 – Present
WorldQuant University, Online, Tuition-Free
Focus: Quantitative Finance, Financial Markets, Financial Data, Python, Statistical Modelling

Non-Degree Studies, Full Stack Web Development Oct 2025
University of Helsinki, Online
Modules: GraphQL, TypeScript, CI/CD, Containers, Relational Databases, React Native, Project Work

B.Eng., Electrical/Electronic Engineering 2021
University of Benin, Nigeria
Focus: Electrical Systems, Electronics, Embedded Systems, Control, Telecommunications

Technical Skills

Programming & Scientific Computing: Python, MATLAB, C/C++, Java, JavaScript/TypeScript, SQL, Bash, R; NumPy, SciPy, pandas, JAX, matplotlib, Plotly, SymPy, python-control.

Control Systems & Modelling: Classical control, state-space modelling, system identification, ARX, ARMAX, Box-Jenkins, Output-Error models, parameter estimation, observers/Kalman filtering, optimization, model validation, residual analysis, digital control, MATLAB/Simulink.

Battery & Energy Systems: Lithium-ion battery modelling, SPMe/DFN concepts, degradation monitoring, electrochemical voltage modelling, current–voltage data analysis, cycle detection, resistance tracking, parameter drift analysis, PyBaMM exposure.

Data Engineering & Analytics: PySpark, Apache Spark, Delta Lake, Databricks, dbt, Airflow/Prefect, Great Expectations, Parquet/Delta, Snowflake SQL, BigQuery SQL, PostgreSQL, MySQL, Power BI, Excel, KPI dashboards, batch and micro-batch pipelines.

Machine Learning & AI: scikit-learn, TensorFlow, PyTorch, Keras, XGBoost/LightGBM, MLflow, feature engineering, model evaluation, time-series modelling, NLP, BERT fine-tuning, RAG, embeddings, vector databases, Streamlit/Dash AI demos.

Software & Web Development: FastAPI, Flask/Django basics, Node.js/Express, React, Next.js, TypeScript, ASP.NET Core, Java Spring Boot, REST APIs, WebSockets, SQLite/PostgreSQL, Swagger/OpenAPI, clean architecture basics.

DevOps, Cloud & Platform: Git/GitHub, GitHub Actions, Jenkins/Travis CI, Docker, Kubernetes basics, Helm, Kustomize, Terraform basics, Linux, venv/conda, AWS basics, Azure Databricks/ADF basics, GCP BigQuery.

Networking & Systems: SNMP, TCP/IP, MPLS exposure, NOC operations, Linux terminal workflows, Bash scripting, tcpdump/Wireshark, UART/I2C/CAN concepts, POSIX threads and IPC exposure.

Professional: Technical writing, teaching, mentoring, Agile/Scrum, stakeholder communication, documentation, experiment logging, reproducible research, analytical problem-solving.

Research and Thesis Project

Physics-Informed Battery Degradation Monitoring using SPMe System Identification M.Sc. Thesis

GitHub Repository

Developed a physics-informed system identification framework to estimate internal lithium-ion battery dynamics and degradation indicators directly from terminal current–voltage measurements. The project combines SPMe-inspired modelling, structured state-space dynamics, nonlinear voltage surrogate learning, and staged parameter identification.

- Designed a structured SPMe-inspired state-space proxy model for lithium-ion battery internal dynamics, including solid diffusion and electrolyte transport behaviour.
- Implemented a staged identification workflow separating nonlinear voltage learning, dynamic model recovery, and joint refinement of physical and surrogate parameters.

- Built nonlinear voltage surrogate models to approximate open-circuit, kinetic, electrolyte, and ohmic voltage contributions from measurable signals and latent states.
- Developed Python/JAX optimization pipelines for parameter estimation, model fitting, cycle-level analysis, and synthetic validation experiments.
- Implemented degradation monitoring metrics including resistance evolution, dynamic parameter drift, and nonlinear voltage-surface drift across cycles.
- Created visualization tools for voltage fits, residuals, degradation trends, learned voltage surfaces, heatmaps, and 3D plots.
- Built a modular research codebase with preprocessing, modelling, system identification, evaluation, visualization, and PyTest-based validation components.

Technologies: Python, JAX, NumPy, SciPy, PyTest, MATLAB, System Identification, Optimization, Battery Modelling, Scientific Computing.

Professional Experience

Graduate Research / Teaching Assistant Jan 2024 – Present
University of Calgary, Schulich School of Engineering Calgary, AB

- Conduct research on physics-informed modelling and system identification methods for lithium-ion battery degradation monitoring using experimental current–voltage data.
- Develop Python-based modelling, preprocessing, parameter-estimation, and visualization pipelines for battery research workflows.
- Build and test modular scientific software using Python, JAX, NumPy, SciPy, PyTest, and Git-based version control.
- Support teaching and student mentoring across courses including analog electronics, applied optimization, software engineering practices, computer organization, control systems, modelling and control of electric machines and drives, and computer architecture/embedded systems.
- Prepare technical documentation, experiment notes, grading rubrics, tutorials, and research updates for academic stakeholders.

Real Estate Assistant May 2024 – Aug 2024
CIR REALTY Calgary, AB

- Supported administrative, documentation, listing, and client-service workflows for a real estate professional.
- Organized documents, assisted with communication, and handled detail-oriented operational tasks requiring accuracy and confidentiality.

Lead Technology Specialist / Network Support Analyst / Network Engineer Intern Nov 2019 – Nov 2023
MTN Nigeria, Network Performance Quality Assurance / Network Operations Lagos, Nigeria

- Supported network performance, troubleshooting, KPI reporting, and operations across NPQA, core performance, NSMC, TX/IP MPLS, data and internet services, and NSS-related functions.
- Used Python, Excel, Power BI, Linux terminal workflows, Bash scripting, and SNMP-related tools to analyze network data, automate recurring tasks, and support performance reporting.
- Monitored network capacity and performance, investigated network faults, supported escalation workflows, and contributed to operational reliability.
- Built dashboards and recurring reports for daily, weekly, and monthly network performance tracking.
- Collaborated with cross-functional technical teams to support fault resolution, network optimization, and process improvement.
- Progressed from network operations/support intern to network engineer intern, network support analyst, and lead technology specialist responsibilities across the full MTN period.

Undergraduate Research Assistant Jan 2020 – Aug 2021
University of Benin, Department of Electrical/Electronic Engineering Benin City, Nigeria

- Conducted embedded systems and IoT research for remote battery monitoring and control applications.
- Prototyped a GSM/GPRS-enabled battery monitoring and control device using microcontroller-based hard-

ware.

- Co-authored research work on remote battery monitoring and IoT-enabled control for energy sustainability.
- ResearchGate

Automation / Electrical Engineer Intern Aug 2019 – Nov 2019
Nigerian Bottling Company, Coca-Cola HBC Nigeria

- Supported troubleshooting, inspection, and maintenance documentation for industrial electrical and automation systems.
- Assisted with work-order tracking, quality checks, and routine support for plant maintenance workflows.

Selected Projects

Battery, Control Systems, and Scientific Computing

Physics-Informed Battery Degradation Monitoring using SPMe System Identification M.Sc. Thesis
GitHub

- Built a physics-informed battery degradation monitoring framework using SPMe-inspired system identification and nonlinear voltage surrogate modelling.
- Implemented cycle detection, signal preprocessing, staged parameter estimation, resistance tracking, parameter drift analysis, and model-visualization tools.

Kalman Filter – 1D Position Tracker Nov 2025
GitHub

- Built an interactive Streamlit demo for a constant-velocity Kalman filter, allowing users to tune process and measurement noise and observe estimation performance.

PLC + HMI + SCADA Simulation Nov 2025
GitHub

- Built a simulated tank-level control process using a Python Modbus TCP PLC simulator, Node-RED dashboard, InfluxDB, and Grafana.

LV Switchboard Approval Package Nov 2025
GitHub

- Prepared a simplified 600 V, 3-phase, 4-wire low-voltage switchboard design package with drawings and schedules for electrical design practice.

Data Engineering, Analytics, and Cloud

Mobility Lakehouse – Polyglot Data Engineering Project
GitHub

- Built a local Dockerized lakehouse using Spark/Scala, Python/Airflow, Great Expectations, Kafka, MinIO, Trino SQL, dbt, Go services, Rust load generation, TypeScript UI, Prometheus, and Grafana.

Olist E-Commerce Medallion Pipeline
GitHub

- Built Bronze/Silver/Gold PySpark and Delta Lake pipelines on the Brazilian Olist dataset, producing curated marts for revenue, delivery SLA KPIs, and customer churn features.

Databricks Mini-Lakehouse
GitHub

- Implemented Scala Structured Streaming to Delta Bronze, Python/PySpark Silver/Gold curation, lightweight data quality checks, BI-ready exports, and MLflow experiment logging.

OSS Lakehouse in Docker
GitHub

- Created a reproducible local lakehouse stack with Spark, Delta Lake, MinIO, MLflow, JupyterLab, PySpark, and Great Expectations.

Supplier Quality & Capability Dashboard

GitHub

- Built a Power BI quality dashboard covering supplier defect rates, Cp/Cpk process capability, NCR aging, and weekly 8D-style triage indicators.

Software Engineering, Backend, and Full Stack

kubeops-fastapi-operator

GitHub

- Built a Kubernetes operator using KOPF to manage FastAPI and PostgreSQL workloads with CRDs, reconciliation logic, rollouts, tests, and runbooks.

Banking System – ASP.NET Core Web API

GitHub

- Built a modular banking-system learning project using ASP.NET Core Web API, Entity Framework Core, SQLite, Razor Pages, Swagger/OpenAPI, xUnit testing, and clean architecture ideas.

Order Dispatch Simulator

GitHub

- Built a FastAPI and Streamlit dispatch simulator comparing nearest, load-balanced, and batched order-assignment strategies using operational metrics.

Order Notifications Demo

GitHub

- Built a real-time order-status demo using FastAPI, WebSockets, React, TypeScript, and Vite.

Tiny Trip Planner

GitHub

- Built a small itinerary-planning app with React, TypeScript, Vite, Leaflet maps, Node.js, and Express.

AI, Machine Learning, and RAG

Cafeteria Menu RAG/LLM Assistant

GitHub

- Built a RAG assistant for cafeteria menus, allergens, and nutrition using FastAPI, Streamlit, PostgreSQL, pgvector, Docker, Sentence-Transformers, and optional OpenAI embeddings.

Clinic Inbox Agent

GitHub

- Built an AI-assisted triage dashboard that classifies clinic-style inbox messages by urgency and category and proposes staff actions.

LLM Pilot Playbook Dashboard

GitHub

- Built a dashboard for explaining LLM pilot trade-offs, including use-case framing, model strategy, latency, cost, and evaluation metrics.

AI for Sales Enablement – Offline Demo

GitHub

- Built an offline rules-based sales-enablement prototype that reads opportunity data, produces pipeline KPIs, scores risk, recommends next steps, and drafts follow-up messages.

E-Commerce Recommender System

GitHub

- Built a Streamlit recommendation system using popularity-based, item-similarity, and cart-aware hybrid recommendation approaches.

Networking, Systems, and Observability

SNMP Network Health Dashboard

GitHub

- Built an SNMP network monitoring dashboard with FastAPI, SQLite, SQLAlchemy, pysnmp, optional SNMP simulator support, and HTML dashboard views.

Cisco-Style Multi-Client TCP Echo Server

GitHub

- Built a BSD-sockets networking project demonstrating TCP sockets, multi-client I/O with poll, line-based protocols, partial reads/writes, and basic testing.

Kubernetes Cost Visibility Lab

GitHub API

- Built a local Kubernetes cost-observability stack using Kind, Kyverno, kube-prometheus-stack, OpenCost, Grafana, FastAPI, Docker, and GHCR.

High-Performance Systems & Algorithms Bench

GitHub

- Built a benchmarking and experimentation suite combining a C++ simulation kernel with a Python experiment runner for repeatable numerical and systems-performance testing.

Testing and QA Automation

API Smoke Tests – reqres.in

GitHub

- Built a pytest and requests-based API smoke test suite with retries, reports, negative tests, and CI-friendly execution.

UI E2E – SauceDemo

GitHub

- Built an end-to-end Selenium and Behave test suite using Gherkin, Page Object Model, and GitHub Actions.

Publications and Research Outputs

Design and Construction of a Remote Battery Monitoring and Control Device Using the Internet of Things

ResearchGate

- Developed and documented an IoT-enabled battery monitoring and control device using microcontroller and GSM/GPRS communication.

Proton Exchange Membrane Electrolyzer Battery Data Prediction Using NARX Neural Network Model

ResearchGate

- Co-authored a predictive modelling study using NARX neural networks for PEM electrolyzer battery voltage forecasting from current and historical voltage data.

Kinematic Analysis of Omnidirectional Mecanum Wheeled Robot

ResearchGate

- Co-authored mecanum robot kinematic analysis work involving wheel-to-chassis mapping and motion behaviour.

Open-Source Contributions

Meshery, Layer5 Dec 2025 – Present

Linux Foundation LFX Mentorship / Open-Source Contribution

- Engaging with the Meshery community through newcomer and working-group activities.

- Reviewing Meshery Server/Operator and Meshery UI repositories while setting up local development, linting, and testing workflows.
- Focus areas include issue triage, bug reproduction, onboarding tasks, small fixes, unit tests, and documentation improvements.

Job Simulations and Professional Development

- **GE Engineering Virtual Experience Program** – Forage, Aug 2022.
- **J.P. Morgan Software Engineering Virtual Experience** – Forage, May 2024.
- **Blackbird Australia Software Engineering Virtual Experience** – Forage, Jun 2024.
- **JPMorgan Chase & Co. Agile Job Simulation** – Forage, Jun 2024.
- **JPMorgan Chase & Co. Software Engineering Lite Virtual Experience** – Forage, Jun 2024.
- **Citi ICG Technology Software Development Job Simulation** – Forage, Jun 2024.
- **Hewlett Packard Enterprise Software Engineering Job Simulation** – Forage, Jul 2024.
- **Lyft Back-End Engineering Job Simulation** – Forage, Jul 2024.
- **Ford EV Engineering Job Simulations** – Forage, Jul 2024.
- **Siemens Mobility Operations Industrial Engineer Job Simulation** – Forage, Aug 2024.
- **Siemens Mobility Project Manager Job Simulation** – Forage, Aug 2024.
- **JPMorgan Chase & Co. Quantitative Research Virtual Experience** – Forage, Sep 2024.

Certifications and Continued Learning

Relevant Completed Learning: Full Stack Web Development modules, software engineering job simulations, agile job simulations, quantitative research simulation, engineering virtual experiences, data analytics and machine-learning guided projects.

Planned: Microsoft DP-203 Data Engineering on Azure; AWS Certified Data Engineer – Associate or AWS Data Analytics track; AWS Certified Machine Learning – Specialty; Google Cloud Professional Data Engineer; ITIL 4 Foundation; Professional Scrum Master I; ROS Developer Certificate or applied robotics/control short course; MathWorks/online MPC or battery modelling short course.

Additional Project Bank

The following projects can be selectively highlighted in role-specific CVs:

- **Synapse-Style SQL on Files** – DuckDB, Spark SQL, Docker, CSV/JSON/Parquet querying.
- **Stock Monitor** – stock trend monitoring and analysis application.
- **Customer Churn Prediction in Telecom** – machine-learning churn analysis project.
- **Data-Driven Insights for Donor Selection** – ML-driven fundraising-campaign analytics.
- **Fine-Tune BERT for Text Classification** – TensorFlow NLP classification.
- **Object Localization with TensorFlow** – CNN-based localization.
- **ASL Recognition with Deep Learning** – CNN image-classification project.
- **Mobile Games A/B Testing with Cookie Cats** – A/B testing and statistical analysis.
- **Predicting Credit Card Approvals** – supervised ML classification.
- **Analyze International Debt Statistics** – SQL analytics project.
- **When Was the Golden Age of Video Games** – SQL joins and set-theory analysis.
- **Arduino Temperature Warning System IoT** – Arduino and thermistor-based monitoring.
- **Bank Account Simulator** – basic programming and application-logic project.
- **Credit Card Validator** – validation utility and programming-practice project.

Professional References

Professional References will be available at request.